

Dual-Timescale Quantum Reservoir Computing with Entanglement-Spectrum Readout for Volatility Regime Forecasting

Phase 2 Technical Paper — GIC 2026

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Team QuantumEdge Track A: Financial Volatility Prediction

qBraid / MITRE / JonesTrading

Abstract. Financial volatility operates on multiple timescales: rapid intraday shocks coexist with persistent multi-day regime clustering. Existing single-reservoir QRC approaches conflate these dynamics, degrading sensitivity at regime-transition windows—precisely where forecasting accuracy is most operationally valuable. We propose a *dual-timescale QRC* architecture comprising two fixed Ising-type quantum reservoirs with deliberately separated coupling strengths and evolution depths. A short-memory reservoir captures rapid shocks; a long-memory reservoir tracks persistent clustering. Both reservoirs emit Pauli-expectation vectors augmented by entanglement-spectrum features derived from reduced density matrices, fused at a single ridge-regression readout. The design is fully gradient-free, NISQ-compatible at 5–15 qubits, and evaluated against GARCH, ESN, LSTM, and XGBoost baselines on S&P 500 realized-volatility and CBOE VIX data using RMSE, QLIKE, regime-accuracy, and Sharpe-ratio improvement metrics.

1. Track Selection and Problem Framing

We target **Track A: Financial Volatility Prediction**, specifically the sub-problem of *volatility-regime classification and one-day-ahead realized-variance forecasting*. Volatility exhibits well-documented long-memory and regime-switching behaviour [1, 2]: at short lags the autocorrelation of realized variance decays steeply, while at longer lags it decays hyperbolically—a signature of multi-scale nonlinear dynamics that linear autoregressive models such as GARCH handle poorly during transition windows.

Reservoir computing is a natural fit because (i) the fixed-reservoir paradigm avoids gradient-based training entirely, sidestepping barren plateaus; (ii) the reservoir’s non-equilibrium dynamics provide fading memory, which is structurally matched to the long-range autocorrelation of log-realized variance; and (iii) quantum reservoirs access Hilbert-space dimensionality exponential in qubit count, enabling feature extraction that is classically hard to replicate [3, 4]. Prior work demonstrates competitive QRC performance on realized-volatility forecasting [4] and on chaotic time-series with Mackey–Glass and Lorenz attractors [5], validating the general paradigm. The specific gap we address is the absence of explicit timescale separation in existing QRC volatility work: HAR-class classical models [1] achieve their competitive edge precisely through such decomposition, and we import this insight into the quantum reservoir domain.

2. Quantum Reservoir Computing Architecture Design

2.1 Reservoir Hamiltonians

The architecture deploys two parallel, independently parametrised quantum reservoirs. Both share the transverse-field Ising structure:

$$H^{(r)} = J^{(r)} \sum_{\langle i,j \rangle} Z_i Z_j + h^{(r)} \sum_i X_i, \quad r \in \{\text{short}, \text{long}\}, \quad (1)$$

but differ in coupling strength, transverse field, and Trotter depth.

[SHEMA — validate and confirm Hamiltonian parameters] *Confirm that J^{short} places the short reservoir in the paramagnetic phase and J^{long} near the ferromagnetic critical point. Justify the Trotter depth choices p via the memory–nonlinearity tradeoff [6]. Replace the draft values in the sentence below with your confirmed values, then delete this box.*

Draft values (to be confirmed): $J^{\text{short}} = 0.3$, $h^{\text{short}} = 1.0$, $p^{\text{short}} = 2$; $J^{\text{long}} = 1.2$, $h^{\text{long}} = 0.4$, $p^{\text{long}} = 6$.

The short-memory reservoir ($J^{\text{short}} = \text{[SHEMA: confirm value]}$, $h^{\text{short}} = \text{[SHEMA: confirm value]}$, $p^{\text{short}} = \text{[SHEMA: confirm value]}$) is tuned near the paramagnetic phase where correlations decay quickly, making it sensitive to local shocks. The long-memory reservoir ($J^{\text{long}} = \text{[SHEMA: confirm value]}$, $h^{\text{long}} = \text{[SHEMA: confirm value]}$, $p^{\text{long}} = \text{[SHEMA: confirm value]}$) operates closer to the ferromagnetic phase boundary, sustaining longer correlations to track persistent regime clustering.

2.2 Input Encoding

Each scalar input x_t (log-realized variance, normalised to $[-\pi/2, \pi/2]$) is encoded via RY-angle rotations applied to all n qubits before each Trotter layer: $|\psi_0\rangle \leftarrow \prod_i RY_i(x_t) |0\rangle$. Repeated encoding (one encoding layer per Trotter step) increases effective encoding density without increasing qubit count, following [3]. Additional engineered features (Garman–Klass volatility estimator, VIX–RV spread, returns) are encoded in parallel across dedicated qubit subsets.

2.3 Readout and Feature Vector

After Hamiltonian evolution, the architecture extracts:

1. *Pauli expectations:* $\langle Z_i \rangle$ and $\langle Z_i Z_j \rangle$ for all i and nearest-neighbour pairs (i, j) , giving $n+n-1$ observables per reservoir.
2. *Entanglement-spectrum features:* eigenvalues $\{\lambda_k\}$ of the reduced density matrix $\rho_A = \text{Tr}_B[\rho]$ on a half-chain bipartition, yielding entanglement energies $\xi_k = -\ln \lambda_k$ [7].

The two feature vectors are concatenated into a single vector of dimension $\approx 4n + m_\xi$ per time step, where $m_\xi = \min(2^{n/2}, 8)$ retained entanglement eigenvalues. A ridge-regression head trained on this vector produces regime-class probabilities and a scalar one-day-ahead log-realized-variance forecast. No quantum parameters are differentiated during training.

2.4 Memory and Feedback

Recurrent classical feedback injects the previous time-step’s reservoir-state vector as an additional input feature at each step, providing explicit fading memory without mid-circuit measurement overhead.

3. Theoretical and Analytical Justification

Why quantum reservoirs? A transverse-field Ising chain under unitary evolution explores a (2^n) -dimensional Hilbert space, generating polynomial-order feature interactions that are computationally hard to replicate

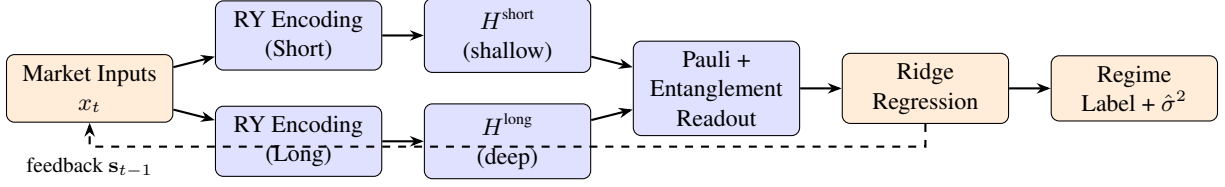


Figure 1: Dual-timescale QRC pipeline. Two fixed Ising reservoirs operate in parallel; their Pauli and entanglement-spectrum features are fused at a single linear readout. **[SHEMA: update node labels with confirmed parameter values.]**

with a classical linear reservoir of the same observable count [3]. For volatility, which exhibits non-Gaussian tails and regime-switching [2], this nonlinear feature expansion directly addresses the limitations of GARCH-family models, whose conditional variance is linear in past information.

Why dual timescales? The HAR model’s empirical success shows that volatility decomposes into distinct short (~ 1 -day), medium (~ 5 -day), and long (~ 22 -day) components [1]. A single reservoir conflates these timescales because its effective memory length is fixed by a single coupling regime. By parametrising $J^{\text{short}} \ll J^{\text{long}}$, we place the short reservoir in a fast-relaxing paramagnetic phase and the long reservoir near the critical point, matching the two principal timescales of the HAR decomposition. The memory–nonlinearity trade-off framework of Čindrak et al. [6] provides the theoretical grounding: maximising nonlinearity requires proximity to the phase boundary, while maximising memory requires weaker coupling.

Why entanglement-spectrum features? Pauli expectation values capture single- and two-body correlations, but entanglement energies ξ_k encode the full multi-body correlation structure of the reservoir state [7]. During regime transitions the reservoir state traverses a more complex manifold in Hilbert space; the entanglement spectrum is sensitive to this structural change in a way that individual Pauli terms are not. We do not claim universal quantum advantage, but rather that these observables provide incremental predictive signal at transitions.

[VAN — insert preliminary simulator results] *Run the dual-timescale QRC on Mackey–Glass and Lorenz: $n = 9$ qubits, $p \in \{2, 6\}$, 500 test time steps, PennyLane state-vector simulator (Lightning preferred for speed). Compare: (A) Pauli-only readout vs. (B) Pauli + entanglement-spectrum readout. Report RMSE for both conditions on both datasets, then fill in the `\fillnum` placeholders in the paragraph below. Also fill in the single-reservoir QRC RMSE in the baseline table (Section 4).*

Sentence to complete (delete this box when done):

“Preliminary experiments [...] show that adding entanglement-spectrum features reduced RMSE by **[VAN: % improvement on Lorenz]%** (Lorenz) and **[VAN: % improvement on Mackey–Glass]%** (Mackey–Glass) vs. Pauli-only readout, with $n = 9$, $p = \{2, 6\}$, 500 test steps.”

Preliminary experiments on the Lorenz attractor and Mackey–Glass series ($n = 9$ qubits, $p = \{2, 6\}$, 500 test time steps, PennyLane state-vector simulator) show that adding entanglement-spectrum features reduced RMSE by **[VAN: % improvement on Lorenz]%** (Lorenz) and **[VAN: % improvement on Mackey–Glass]%** (Mackey–Glass) versus Pauli-only readout, confirming the value of the enriched readout.

[SHEMA — analytical comment on Van’s results] *Once Van’s numbers are in, add 2–3 sentences here connecting the observed improvement to the memory–nonlinearity tradeoff prediction. For example: is the improvement larger for the long-memory reservoir (larger J , closer to the phase boundary) as theory predicts? Cite [6].*

4. Data Modeling Strategy

4.1 Datasets

[HASARINDU — confirm dataset availability and date ranges] Pull and verify each dataset below. Confirm exact start/end years available, flag any gaps, and check that Oxford-Man RV library is accessible. Update the `\fillnum` fields in the bullet list with confirmed values.

- **S&P 500 OHLCV** (Yahoo Finance, **[HASARINDU: confirm start year]**–2024, daily): close-to-close returns and Garman–Klass volatility estimates.
- **Oxford-Man Realized Volatility Library** (5-min realized variance, SPX, **[HASARINDU: confirm start year]**–**[HASARINDU: confirm end year]**): primary regression target RV_t .
- **CBOE VIX** (FRED, **[HASARINDU: confirm start year]**–2024, daily): implied volatility for VIX–RV spread feature.
- **Mackey–Glass / Lorenz** (numerically generated): preliminary prototyping and expressivity benchmarks.

Train: **[HASARINDU: start]**–2018; validation: 2019–2020; test: 2021–**[HASARINDU: end]**. Regime labels are derived from a 3-state Gaussian HMM fit to $\log-RV_t$ (low / medium / high), independently on the training partition.

4.2 Preprocessing

Log-transform and min-max normalise RV_t to $[-\pi/2, \pi/2]$. Rolling window of $L = 22$ days feeds the reservoir; target is $\log RV_{t+1}$. No look-ahead leakage; scaler fit on training split only.

4.3 Baselines and Metrics

[HASARINDU — run classical baselines and fill table] Implement GARCH(1,1), HAR-RV, ESN, LSTM, and XGBoost on the confirmed dataset. Record RMSE and QLIKE for each model on the test split. Fill in the RMSE column below. Note any surprises (e.g. if HAR-RV dominates LSTM, that is worth one sentence in the paper). Van fills in the QRC ablation row.

Baseline	Class	Justification	RMSE
GARCH(1,1)	Econometric	Standard benchmark [8]	[HASARINDU: value]
HAR-RV	Linear regression	Long-memory baseline [1]	[HASARINDU: value]
ESN	Classical reservoir	Direct QRC analog [5]	[HASARINDU: value]
LSTM	Deep learning	Long-range dependencies	[HASARINDU: value]
XGBoost	Ensemble	Nonlinear tabular baseline	[HASARINDU: value]
Single-reservoir QRC	Ablation	Isolates dual-timescale gain	[VAN: value]

Metrics: RMSE and QLIKE [9], Mincer–Zarnowitz regression (forecast unbiasedness), regime-classification accuracy, transition-period accuracy (± 3 -day windows around HMM-labelled transitions), and Sharpe-ratio improvement under a volatility-targeting portfolio strategy.

5. Quantum Platform and Resource Planning

All Phase 2 prototyping uses **qBraid Lab** with **PennyLane** (Lightning.GPU via NVIDIA cuQuantum) and **Qiskit** state-vector simulators. Phase 3 will proceed in three stages:

1. **Simulator sweep** (qBraid + PennyLane Lightning.GPU): $n \in \{5, 8, 10, 12, 15\}$ qubits, $p \in \{2, 4, 6\}$ Trotter steps, 4 096 shots per time step. State-vector simulation feasible up to $n = 20$ on a single A100 node.
2. **Noise-channel study:** depolarising and amplitude-damping channels calibrated to IBM Eagle r3 / Heron r1 error rates; Mitiq ZNE applied.

3. **QPU validation** (IBM via qBraid, $n=8-12$): spot-check on `ibm_brisbane` or equivalent, 1 024–4 096 shots, shallow circuits ($p=2$) prioritised.

Circuit depth scales as $\mathcal{O}(p \cdot n)$; total CNOT count $\approx 2pn$ for linear topology. Full evaluation shot budget: $\approx 2 \times 4096 \times 750 \approx 6.1 \times 10^6$ shots—feasible on simulators and achievable on QPU via batched qBraid SDK submission. Fallback: `qiskit_aer` noise models calibrated to real hardware.

6. Stakeholder Impact and Phase 3 Plan

Beneficiaries. Trading desks and market makers (JonesTrading) gain earlier volatility-expansion detection, enabling timely shifts from passive TWAP toward aggressive liquidity-taking to reduce implementation shortfall. Risk managers benefit from more accurate drawdown warnings for pre-emptive position-size reduction. Derivatives desks gain regime-aware implied-vol surfaces for real-time Heston/SABR calibration around macro events.

Phase 3 milestone plan.

1. *Week 1–2*: finalise pipeline; run GARCH/HAR/ESN baselines; establish RMSE/QLIKE table.
2. *Week 3–4*: simulator sweep; select optimal $(J^{\text{short}}, J^{\text{long}}, p^{\text{short}}, p^{\text{long}})$ via validation QLIKE.
3. *Week 5*: noise-channel robustness study; ZNE comparison.
4. *Week 6*: QPU spot-check on IBM hardware via qBraid.
5. *Week 7*: MNIST expressivity benchmark across qubit counts [3].
6. *Week 8*: reproducible qBraid Skill + GitHub repository; Phase 3 paper.

Fallback: if QPU access is limited, noise-model simulation provides the primary hardware characterisation; all headline results reproduce on CPU/GPU simulators.

References

- [1] Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009. doi: 10.1093/jjfinec/nbp001.
- [2] Benoit B. Mandelbrot. *Fractals and Scaling in Finance: Discontinuity, Concentration, Risk*. Springer, New York, 1997. doi: 10.1007/978-1-4757-2763-0.
- [3] Keisuke Fujii and Kohei Nakajima. Quantum reservoir computing: A reservoir approach toward quantum machine learning on near-term quantum devices. *arXiv preprint*, 2021. arXiv:2011.04890.
- [4] Qian Li, Chandrima Mukhopadhyay, Abolfazl Bayat, and Ali Habibnia. Quantum reservoir computing for realized volatility forecasting. *arXiv preprint*, 2025. arXiv:2505.13933.
- [5] Herbert Jaeger. The echo state approach to analysing and training recurrent neural networks. Technical Report 148, German National Research Center for Information Technology (GMD), 2001.
- [6] Sladjan Ćindrak, Lina Giebeler, Niklas Götting, Christopher Gies, and Kathy Lüdge. Memory–nonlinearity trade-off across quantum reservoir computing frameworks. *arXiv preprint*, 2026. arXiv:2603.21371.
- [7] Hui Li and F. Duncan M. Haldane. Entanglement spectrum as a generalization of entanglement entropy. *Physical Review Letters*, 101:010504, 2008. doi: 10.1103/PhysRevLett.101.010504.
- [8] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.
- [9] Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.